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PALAEOMAGNETISM AND K-Ar AGE OF MESOZOIC AND CENOZOIC IGNEOUS ROCKS FROM ANTARCTICA

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A new analysis of palaeomagnetic data for igneous rocks from Deception Island, 25 de Mayo Island (King George Island) and Cape Spring, are given

K-Ar age determinations indicate that most of the igneous samples from 25 de Mayo Island included in the palaeomagnetic study are of Late Mesozoic/Early Tertiary age The significance of these palaeomagnetic-radiometric data on the hypothesis of oroclinal bending of the Antarctic Peninsula and on the apparent polar movement of Antarctica is discussed

The positions of palaeomagnetic poles for the Andean igneous complex indicate that there has not been any apparent post-Late Cretaceous/Early Tertiary oroclinal bending in the Antarctic Peninsula from 74°S to 62°S

A comparison of the positions of palaeomagnetic poles for Antarctica and Australia suggests that the direction of apparent polar movement relative to Antarctica reversed after the Miocene

1 Introduction

Palaeomagnetic studies of igneous rocks from the 25 de Mayo (King George Island, Miocene) and Deception (Plio-Pleistocene) Islands (South Shetland Group) and Graham Land (Cape Spring, Late Cretaceous to Early Tertiary) were carried out Details of the field and laboratory techniques used in these studies have been published elsewhere [1,2] In this paper we present a new analysis of the palaeomagnetic data obtained in these studies and a discussion of the implications of all these data on the dynamical history of the Antarctic Peninsula and the apparent polar path of Antarctica K-Ar age determinations were carried out on some of the igneous rocks from the 25 de Mayo Island, the K-Ar ages suggest an older age for these igneous rocks than previously supposed The significance of this on the dynamics of the Antarctic Peninsula is also discussed

2 Geological evidence and sampling

Deception Island Situated between 62°53′, and 63°02′S and 60°30′ and 60°45′W, the island is an active volcano which has erupted lavas during different magmatic episodes since Plio-Pleistocene times The oldest lava flows have been grouped in the Fumarola Bay Volcanics [3] and the youngest in the Volcánica Antigua and Volcánica Moderna Series [4] which do not include the lava flows erupted in recent times

The Volcánica Antigua and Volcánica Moderna Series were sampled at ten different sites (two andesites and eight basalts, ten hand samples) during the field work carried out in 1966–1967 by the Instituto Antartico Argentino The palaeomagnetic study of the Fumarola Bay Volcanics was carried out by Blundell [5]

25 de Mayo Island In the western part of the island (between 62°11′ and 62°14′S, 58°53′ and 59°00′W) andesites, basalts and tuffs crop out Schauer and Fourcade [6] have suggested a Late Miocene age for these igneous rocks and Orlando [7] has studied a flora enclosed in the tuffs and regards it as being close to the Lower/Middle Miocene boundary

Whole K-Ar age determinations were carried out on four of the samples from 25 de Mayo Island used for the palaeomagnetic study (Table 1) The study was carried out in the Instituto de Geocronología y Geología Isotópica (Universidad de Buenos Aires) A petrological analysis indicated that the samples do not show significant alteration Three of the samples (A23, A11 and A24) yield ages older than Miocene (Early Cretaceous to Early Palaeocene) This Late Mesozoic/Early Tertiary age for the volcanic rocks exposed in 25 de Mayo Island is consistent with the Late Mesozoic age of volcanic rocks found in Livingston Island (South Shetland Group) by Dalziel and Elliot [8] On the basis of the geologic map [1] it seems to us that most of the volcanic samples from 25 de Mayo Island included in the palaeomagnetic study (section 3) are of Late Mesozoic/Early Tertiary age (Andean igneous complex) Twenty-eight hand samples were collected (27 andesites and 1 basalt) during the field work carried out in 1966–1967

Cape Spring In Cape Spring (Danco's Coast, Antarctic Peninsula, 64°S 62°W) rhyodacitic rocks of Late Cretaceous to Early Tertiary age, which have been assigned to the Andean igneous complex [9], are exposed Twenty-two hand samples were collected from five different rhyodacitic rock units in this area ten were collected from a lava flow (Cerro Escombrera) and twelve from four dikes which cut granite porphyry exposures (Andean intrusives) at the northeast of Cape Spring

Whole K-Ar age determinations yield an age of 94 ± 6 m y for the lava flow from Cerro Escombrera and 102 ± 5 m y for one of the dikes [2]

3 The palaeomagnetic study

Four cylinders of 2 5 cm diameter and height were cut from each hand sample Detailed alternating field demagnetization was used to isolate the stable remanence of one pilot cylinder of each sample (optimum alternating field) Two other cylinders were submitted to this optimum alternating field so as to improve the definition of the direction of the stable remanence of each hand sample One other cylinder was submitted to detailed thermal cleaning

Deception Island One hand sample was destroyed during transportation The other nine samples have very stable natural remanent magnetization (NRM) The optimum demagnetizing AC field varied between 100 and 150 Oe at which more than 75% of the NRM intensity was retained The blocking temperatures cover the range from 200 to 400°C The mean directions of cleaned remanent magnetization [10] of each hand sample are shown in Fig 1

The direction of the stable remanence of sample D6M points to the south and downward, the other eight samples have normal cleaned remanence The figure shows that the directions of the stable remanence of the normal samples are closer to the direction of the axial dipole than to that of the pres-

TABLE 1

K-Ar ages for the andesitic lava flows from 25 de Mayo Island

Sample No	Rock	Site	NRM polarity *	K (%)	$^{40}Ar_{rad}$ (10^{-10} mol/g)	$^{40}Ar_{atm}$ (%)	Age (m y)
A6	andesite	Ardley Island	N	0 30	0 140	68	27 ± 2
A24	andesite	East Flat Top Point	N	0 28	0 302	23	61 ± 3
A11	andesite	Northeast Flat Top Point	R	0 54	0 843	61	88 ± 5
A23	andesite	Fildes Strait	R	0 20	0 391	84	110 ± 10

* N = normal, R = reversed

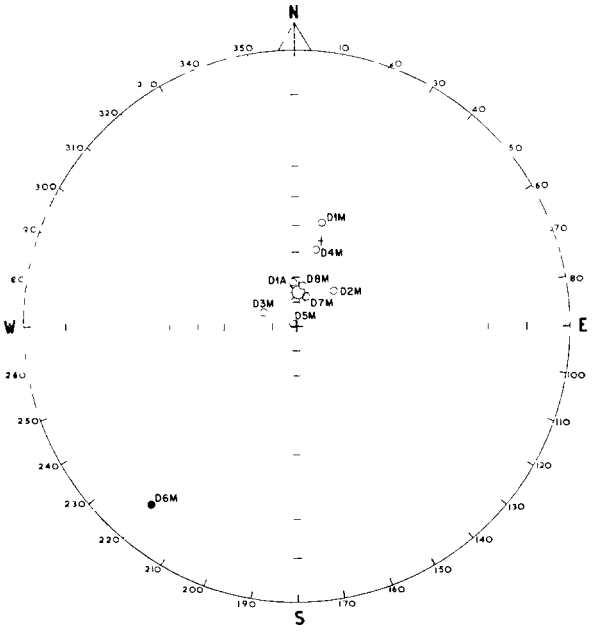


Fig 1 Deception Island Mean directions of cleaned remanent magnetization of hand samples Solid symbols indicate downward dipping directions (lower hemisphere) The directions of the present geomagnetic field and of the axial dipole are shown by + and ✲, respectively

ent geomagnetic field in the zone To proceed further a 40° cut-off was used to classify the stable remanence of lava flows [11,12], we use the criterion that poles with VGP co-latitudes >40° are intermediate between normal and reversed For this, virtual geomagnetic poles for each lava flow were calculated, a palaeomagnetic pole was then calculated by meaning these virtual geomagnetic poles We rejected VGP's situated more than 40° from the palaeomagnetic pole and a new mean pole was computed In this way a population of virtual geomagnetic poles was obtained all within 40° of the palaeomagnetic pole, this population gives the mean pole at 86°S, 158°E ($N = 8, k = 11, \alpha_{95} = 17°$) The rejected VGP is classified as oblique (D6M, oblique, near reversed)

25 de Mayo Island Two hand samples were destroyed during transportation, it was not possible to isolate stable remanence in four samples The optimum demagnetizing AC field for the other twenty-two samples varied between 100 and 150 Oe at which more than 30% of the NRM was retained in the majority of them The blocking temperatures cover the range from 450 to 550°C for the majority of samples The mean directions of the stable remanent magnetization of each hand sample are shown in Fig 2 The directions of stable remanence of ten samples are downward (reversed or oblique near reversed), the directions of the other samples are upward (normal or near normal) Fig 2 shows that these directions are scattered and not antiparallel That may reflect a real behaviour of the geomagnetic field or indicate that the primary remanence has not been properly isolated As four disks of each sample were submitted to AC or thermal cleaning to isolate the primary remanence, we think that it reflects the behaviour of the geomagnetic field during the cooling of lava flows Fig 2 shows that the directions of stable remanence are closer to the directions of the axial dipole than to that of the present geomagnetic field To proceed further, these directions were classified as normal, reversed or oblique in the same way as before Seven samples have normal cleaned remanence, four samples have reversed remanence, the other eleven have oblique remanence (six oblique near normal, five

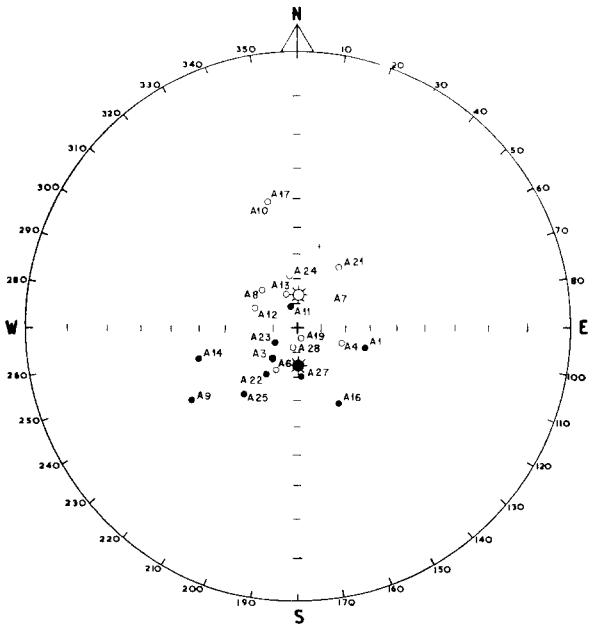


Fig 2 25 de Mayo Island Mean directions of cleaned remanent magnetization of samples The key to the symbols is as given in Fig 1

oblique near reversed) Blundell [5] found only normal remanence in samples from this island The large number of oblique samples suggests that the lava flows from 25 de Mayo Island were extruded during a time span of frequent reversals of the geomagnetic field in the Late Cretaceous/Early Tertiary

The mean direction of the stable remanence of normal and reversed samples is given in Table 2 The population of VGP's, all within 40° of the mean pole, gives the mean pole at 85°S, 200°E ($N = 11, k = 10, \alpha_{95} = 15°$)

Cape Spring The optimum demagnetizing AC field for the samples from Cerro Escombrera was 200 Oe at which more than 80% of the NRM intensity was retained, the best demagnetizing field for the samples form the four dikes varied between 200 and 250 Oe at which more than 60% of the NRM was retained The blocking temperatures cover the range from 540 to 580°C The mean directions of the stable remanent magnetization of each rhyodacitic unit are given in Table 3, all the units have normal cleaned remanence The mean direction of the stable remanence of these units is given in Table 2 The coordinates of the virtual geomagnetic poles computed for the five igneous units are summarized in Table 3, the mean of these VGP's is 86°S, 117°E ($N = 5, k = 77, \alpha_{95} = 9°$)

4 Interpretation of results

4 1 The age of vulcanism in Deception Island

Blundell [5] found only normal remanence in seven lava flows assigned to the Fumarola Bay Volcanics, Neptunes Bellows Group and the Pendulum Cove Group We found normal (eight sites) and oblique near reversed (one site) cleaned remanence in the Volcánica Antigua and Volcánica Moderna Series This indicates that the geomagnetic field was predominantly of normal polarity during the time of extrusion of those lava flows This, given the age assigned to the lava flows, suggests that the majority of the lava flows of Deception Island were extruded during the Brunhes Normal Epoch (0–0 7 m y), if this is the case, the oblique near reversed remanence isolated in one of the samples indicates that this flow was extruded during either a excursion or a transition of

TABLE 2

Palaeomagnetic data for the Andean igneous complex exposures along the Antarctic Peninsula and offshore islands

Locality	Approximate location	Mean cleaned remanent magnetization						Palaeomagnetic pole				Name	Reference
		N	$D(°)$	$I(°)$	k	$\alpha_{95}(°)$	$\delta_{95}(°)$	lat , long	$d\psi(°)$	$\alpha_{95}(°)$	$d\chi(°)$		
25 de Mayo Island	62°12'S, 59°W	11	10	−74	28	9	35	85°S, 192°E	15		16	AnK_5	–
Livingston Island	62°40'S, 60°W	4	337	−72	110	9	30	77°S, 48°E	13		15	AnK_4	[13]
	63°S, 57°W	4	358	−74	404	5	18	87°S, 107°E	8		9	AnK_1	[13]
Cape Spring	64°S, 62°W	5	0	−73	199	5	17	86°S, 118°E	8		9	AnK_3	–
	65°S, 64°W	6	352	−78	135	6	12	86°S, 172°E	11		11	AnK_2	[13]
Lassiter Coast	73–74°S, 63°W	17	0	−82	33	6	49	87°S, 229°E		11 5		AnK_6	[15]

TABLE 3

Summary of palaeomagnetic data for the Cretaceous rhyodacitic units from Cape Spring, Antarctic Peninsula

Rock unit	K-Ar age (m y)	Mean cleaned remanent magnetization					Virtual geomagnetic pole			
		N	$D(°)$	$I(°)$	k	$\alpha_{95}(°)$	lat (°S)	long (°E)	$d\psi(°)$	$d\chi(°)$
Co Escombrera	94 ± 6	10	358	−76	57	6	89	57	8	9
Dike	–	3	4	−73	1402	3	84	138	5	6
Dike	102 ± 5	3	18	−75	935	4	82	206	7	7
Dike	–	3	354	−65	139	11	73	104	14	17
Dike	–	3	347	−77	183	9	84	4	16	17

the geomagnetic field within the Brunhes Normal Epoch However, we cannot rule out a late Matuyama age for it The mean direction of the stable remanence of the normal sites in Deception Island is $D = 11°$ $I = -73°$ ($N = 15, k = 46, \alpha_{95} = 6°$)

The population of VGP's for the Fumarola Bay Volcanics, Neptune Bellows and Pendulum Cove Groups [5] and the Volcánica Antigua and Volcánica Moderna Series, all within 40° of the mean pole, gives a palaeomagnetic pole for the Deception Island at 85°S, 191°E ($N = 15, k = 19, \alpha_{95} = 9°$, AnQ_1)

4 2 Test for oroclinal bending of the Antarctic Peninsula since the Andean orogeny

The Andean intrusives were formed during the Late Cretaceous/Early Tertiary Andean orogeny A number of authors have indicated the similarity in the geology of the Andes and the Antarctic Peninsula Some of them have postulated that an originally continuous and rectilinear Andean-Antarctandean Cordillera has been bent and fragmented in the vicinity of the Drake Passage to form the Scotia arc (quoted in Dalziel et al [13]) The similarity of the Early Cretacous rocks on South Georgia Island to those on Navarino Island (southern Tierra del Fuego) suggests that disruption and fragmentation of the original cordillera occurred after the Andean orogeny (after the latest Cretaceous and before the Mid-Cenozoic [13]), however, the oroclinal bending of this cordillera may have started during the Andean orogeny Therefore, palaeomagnetic data for samples of the Andean igneous complex and older rocks might keep a memory of any secondary distortion of the original cordillera and be used as a test of the oroclinе hypothesis This was the idea of Dalziel et al [13] when they carried out the palaeomagnetic study of samples from the Andean intrusives and dikes cutting these intrusives

On basis of palaeomagnetic data Hamilton [14] suggested a clockwise swing (25°) along the Antarctic Peninsula from latitude 68°S northeastward since Cretaceous time and, on the contrary, Dalziel et al [13] indicated that the shape of the peninsula from 68°S to 63°30′S has not changed significantly since Andean intrusives were emplaced We shall compare the magnetization of the Andean igneous complex from Cape Spring and 25 de Mayo (given in this paper) and from Lassiter Coast (recently presented by Kellogg and Reynolds [15]) with the palaeomagnetic data used by those authors as a test of the oroclinе hypothesis in the Antarctic Peninsula

Blundell [5] presented palaeomagnetic data for samples collected from Andean intrusive exposures spaced along the Antarctic Peninsula and offshore islands Dalziel et al [13] grouped these data according to the latitudes of the sampling sites and recalculated the mean direction of remanent magnetization, these data are given in Table 3 (latitudes 63° and 65°S) In this table the mean magnetizations of the Andean suite from Livingston Island (62°40′S [13]), Cape Spring (64°S, 102 and 94 m y), Lassiter Coast (73–74°S, 119 m y to 95 m y) and 25 de Mayo Island (62°12′S, 110 to 61 m y and 27 m y) are also given An examination of the table shows that (1) the Andean exposures from the Antarctic Peninsula have inclinations comparable to that of the axial dipole field (−76° for the northern sites, 82° for Lassiter), and (2) their mean declinations do not show a systematic variation with latitude, this sug-

gests that the shape of the Antarctic Peninsula would not have changed significantly since the Andean suite was emplaced However, before accepting this interpretation we should analyze the palaeomagnetic data from other points of view

Firstly, we should consider the uncertainty parameter for declination ($\delta_{95} = \sin^{-1}$ [$\sin \alpha_{95}/\cos I$] [15]) for each site (Table 2), this represents the confidence limit at the 95% level of the mean declination This parameter should be less than the rotation suggested for the Antarctic Peninsula to be used as a test of the oroclinal hypothesis Table 2 shows that δ_{95} parameters for Lassiter Coast, Livingston Island and 25 de Mayo Island are large and the mean declinations for these sites should not be used for testing the hypothesis However, we should note that the large confidence limits for Lassiter Coast are the results of nearly vertical mean inclinations [15] The palaeomagnetic data from this locality are particularly important because they come from collecting sites situated in the southern Antarctic Peninsula, where the oroclinal bending is minimal according to Hamilton [14] and the reconstruction of Dalziel et al [13] Within the confidence limits of the three acceptable sites no oroclinal bending can be unambiguously discerned between 63°S and 65°S

Secondly, we should consider the classical method of comparison of palaeomagnetic pole positions Palaeomagnetic poles for each sampling site on the Antarctic Peninsula were calculated (Table 2, Fig 3) This figure shows that the position of the palaeomagnetic pole for the Livingston Island tonalite and basic dikes (AnK_4 [13]) is not close to the other five Andean igneous complex poles This may be caused either by an inadequate bedding plane correction of palaeomagnetic data or by local rotation of small lithospheric blocks If the first statement is the case we should not use the palaeomagnetic data of this island for tectonic interpretations, if the second is the case, the Livingston Island is the only collecting site of the Antarctic Peninsula which would have been submitted to significant local rotations since the Andean orogeny

Fig 3 shows that the positions of the other five palaeomagnetic poles are reasonably well aligned with the axis along the sampling sites This indicates that local relative rotations have not occurred between the collecting sites along the Antarctic Peninsula from

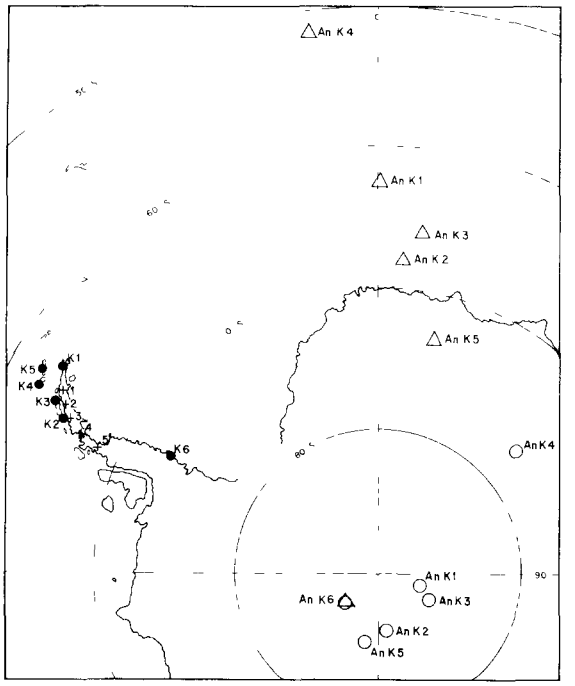


Fig 3 Late Cretaceous/Early Tertiary palaeomagnetic poles for Antarctica (○, references in Table 2) and portion of Antarctica showing sampling sites (●) and rotation poles (+) used to straighten out the Antarctic Peninsula as suggested by Dalziel et al [13] Pole positions after straightening are also shown (△)

where these poles come from The positions of palaeomagnetic poles should be scattered along a direction perpendicular to that axis if local rotations had occurred

Finally we have carried out a palaeomagnetic test of the form of the Antarctic Peninsula suggested by Dalziel et al ([13], Fig 1) valid for Late Cretaceous/Early Tertiary time To do this, we have divided the northern Antarctic Peninsula into five blocks and have used five poles of rotation (indicated by crosses in Fig 3, rotation angles from south to north 17 5°, 15 7°, 15 7°, 8 5° and 7 7°) to straighten it out To recalculate the after straightening positions of palaeomagnetic poles (AnK_1, AnK_4, AnK_5) from collecting sites situated in the northernmost of these blocks (K_1, K_4, K_5) we retate them first using pole of rotation 1, the new palaeomagnetic pole positions obtained were then rotated using pole of rotation 2 and the process was repeated up to pole of rotation 5 In

the same way palaeomagnetic pole AnK_3 was rotated about rotation poles 2, 3, 4 and 5 and palaeomagnetic pole AnK_2 was rotated about rotation poles 3, 4 and 5 The recalculated positions of palaeomagnetic poles after these rotations are indicated by triangles in Fig 3 The scatter of pole positions is increased by straightening out the peninsula the mean of the five palaeomagnetic poles before straightening is 87°S, 165°E (k = 741, α_{95} = 2 8°, AnK) and after 72°S, 1°E (k = 47, α_{95} = 11 2°) If we consider the six poles the figures are 88°S, 110°E (k = 156, α_{95} = 5°) and 69°S, 1°E (k = 35, α_{95} = 11°), respectively Therefore, palaeomagnetic poles from localities situated from 74°S to 62°S support the contention that there has not been any apparent post-Andean orogeny oroclinal bending in the Antarctic Peninsula

We think that AnK is the more representative pole position for the Andean igneous complex exposures at Antarctic Peninsula This indicates that the Antarctic Peninsula occupied latitudes close to its present ones in Late Cretaceous/Early Tertiary times

4 3 General discussion

Turnbull [16] computed a palaeomagnetic pole (81°S, 94°E, $d\psi$ = 7°, $d\chi$ = 8°, AnT_1) for Cenozoic igneous rocks exposed close to Cape Hallet (72°S, 171°E), an age older than Brunhes has been suggested for these rocks [1]

Blundell [5] presented palaeomagnetic data for the Middle Miocene James Ross Island Volcanic Group (sampling site 64°S, 58°W), these data yield a palaeomagnetic pole at 82°S 127°E ($d\psi$ = 20°, $d\chi$ = 23°, AnT_2)

Fig 4 shows the position of the Late Cretaceous and Cenozoic palaeomagnetic poles for Antarctica and Australia [17] on the reconstruction of the Australian-Antarctic plate obtained from sea-floor spreading data [18] The position of Andean igneous complex palaeomagnetic pole is close to the position of the Cretaceous pole for Australia (K(93)) This indicates that the reconstruction of Fig 4 is valid for Cretaceous time The sea-floor spreading data suggest that the separation of those continents occurred in Eocene/Oligocene times (about 40 m y) The different positions of the Tertiary poles for Australia and Antarctica (Fig 4) supports this Therefore, evidence suggests that Australia and Antarctica had a common

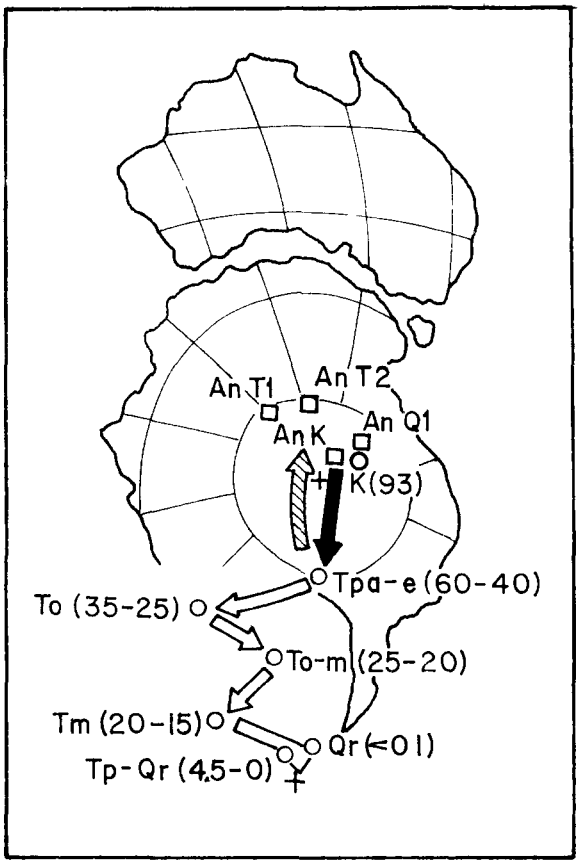


Fig 4 Late Cretaceous and Cenozoic palaeomagnetic poles for Antarctica (□, this paper) and Australia (○, [15]) on the reconstruction of the Australian-Antarctic plate suggested by sea-floor spreading data Filled arrow indicates common polar wander path for the Australian-Antarctic plate, open arrows, polar path for Australia, dashed arrow, polar path for Antarctica The numbers within brackets indicate the age assigned to the poles

apparent polar path in Late Cretaceous/Early Cenozoic times (that defined by the poles AnK-K(93) and Tpa-e (60-40) for Australia, black curve, in Fig 4) The positions of the Cenozoic poles close to the Late Cretaceous poles for Antarctica may be explained if this continent reversed its apparent polar movement after the Miocene The first stage of this movement is that defined by the Late Cretaceous/Early Cenozoic common polar path for the Australian-Antarctic plate in Fig 4 The reversed movement is that indicated by the striped curve and started after the fragmentation of the Australian-Antarctic plate

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